0.1 Setup: generating functions

Encode a multiset S by the polynomial $f(x) = \sum_{s\in S} x^s$, where the coefficient of x^k is the multiplicity of k. Since the elements lie in $\{1,\dots,19\}$, the polynomial f has nonnegative integer coefficients, $f(0)=0$, and $\deg f = \max(S)$.

A partition $S = A_i \sqcup B_i$ with $A_i = i + B_i$ corresponds exactly to writing

$$f(x) = g_i(x) + x^i g_i(x) = (1+x^i)\, g_i(x),$$

where g_i (the generating function of B_i) has nonnegative integer coefficients. Thus:

> S has index $\ge n \iff$ for every $i \in \{1,\dots,n\}$, $f = (1+x^i)g_i$ with $g_i \ge 0$ (all coefficients ≥ 0).

0.2 A divisibility obstruction

Lemma 1. If S has index $\ge n$, then f is divisible in $\mathbb{Z}[x]$ by $L_n:=\operatorname{lcm}*{1\le i\le n}(1+x^{i})$, and $$L_n=\prod*{\substack{d\text{ even}\ d\le 2n}}\Phi_d(x),\qquad \deg L_n=\sum_{\substack{d\text{ even}\ d\le 2n}}\varphi(d),$$ where Φ_d is the d-th cyclotomic polynomial.

Proof. Each $g_i \in \mathbb{Z}[x]$, so $1+x^i \mid f$ in $\mathbb{Z}[x]$ for all $i \le n$. Using $1+x^i = \frac{x^{2i}-1}{x^i-1} = \prod_{d\mid 2i,\, d\nmid i} \Phi_d$, one checks $\Phi_d \mid 1+x^i$ requires d even, and for even d the least such i is $i = d/2$. Hence the cyclotomic factors appearing among $\{1+x^i : i \le n\}$ are exactly Φ_d with d even, $d \le 2n$. These are distinct irreducibles each dividing f, so their product L_n divides f. □

Computing $\sum_{d \text{ even} \le 2n} \varphi(d)$:

n	1	2	3	4	5	6	7
$\deg L_n$	1	3	5	9	13	17	23

Since $\deg f = \max(S) \le 19$, Lemma 1 with $\deg L_7 = 23 > 19$ gives **index** ≤ 6.

0.3 Ruling out index 6

Suppose index ≥ 6. Then $L_6 \mid f$ with $\deg L_6 = 17$, and $f(0)=0$ forces $x \mid f$; as $\gcd(L_6, x) = 1$ we get $f = L_6\, q$ with $q(0) = 0$ and $\deg q = \deg f - 17 \le 2$, so $q = ax + bx^2$.

Now $g_1 = f/(1+x) = R\, q$ where $R := L_6/(1+x) = \Phi_4\Phi_6\Phi_8\Phi_{10}\Phi_{12}$. The low-order coefficients of R are $r_0 = 1,\ r_1 = -2,\ r_2 = 3$ (from Φ_6, Φ_{10} contributing $-x$ each). Requiring $g_1 \ge 0$:

$$[x^2]:\ -2a + b \ge 0, \qquad [x^3]:\ 3a - 2b \ge 0, \qquad [x^1]:\ a \ge 0.$$

These give $2a \le b \le \frac{3}{2}a$, hence $a \le 0$, so $a = b = 0$. Thus no nonzero f exists: **index 6 is impossible.**

0.4 $N = 5$ is attained

Take $f(x) = x\prod_{i=1}^{5}(1+x^i)$. For each $i \le 5$, $g_i = x\prod_{j\ne i}(1+x^j) \ge 0$, so the index is ≥ 5. Since $f = x(1+x)^2\,\Phi_2\Phi_4\Phi_6\Phi_8\Phi_{10}$ contains no factor Φ_{12}, we have $1 + x^6 \nmid f$, so the index is exactly 5. Its maximum element is $16 \le 19$. Hence

$$\boxed{N = 5.}$$

0.5 The minimal maximum element M

For index 5, Lemma 1 gives $L_5 \mid f$ with $L_5 = \Phi_2\Phi_4\Phi_6\Phi_8\Phi_{10}$, $\deg L_5 = 13$; together with $x \mid f$, we get $f = L_5\, q$, $q(0) = 0$, so $\deg f = 13 + \deg q$ with $\deg q \geq 1$.

The key quotient is $g_1 = R_1\, q$ with $R_1 := L_5/(1 + x) = \Phi_4\Phi_6\Phi_8\Phi_{10}$, whose coefficients are

$$R_1 = [\,1, -2, 4, -5, 7, -7, 8, -7, 7, -5, 4, -2, 1\,].$$

- $\deg q = 1$ ($q = q_1x$): $g_1 = q_1xR_1$ inherits the negative coefficients of R_1 — impossible.
- $\deg q = 2$ ($q = q_1x + q_2x^2$): the coefficients $[x^2], [x^3]$ of g_1 force $q_2 = 2q_1$, then $[x^5] = 7q_1 - 5q_2 = -3q_1 \geq 0$ forces $q_1 = q_2 = 0$ — impossible.

So $\deg q \geq 3$, i.e. $\deg f \geq 16$, giving $\max(S) \geq 16$. The set above attains 16, so

$$\boxed{M = 16.}$$

0.6 Minimal number of elements

Now $f = L_5\, q$ with $q = q_1x + q_2x^2 + q_3x^3$, $q_3 \geq 1$ (leading coefficient positive). The condition $g_1 = R_1q \geq 0$ gives, reading off coefficients of g_1:

$$[x^1]:\ q_1 \geq 0, \qquad [x^2]:\ q_2 \geq 2q_1, \qquad [x^{15}]:\ q_3 \geq 0.$$

Also $q_1 = 0$ is impossible: then $[x^3]:\ q_3 \geq 2q_2$ and $[x^{14}]:\ q_2 \geq 2q_3$ give $q_3 \geq 4q_3$, so $q_3 = 0$. Hence $q_1 \geq 1$, and therefore $q_2 \geq 2q_1 \geq 2$, $q_3 \geq 1$, so

$$q(1) = q_1 + q_2 + q_3 \geq 4.$$

Since $L_5(1) = \Phi_2(1)\Phi_4(1)\Phi_6(1)\Phi_8(1)\Phi_{10}(1) = 2 \cdot 2 \cdot 1 \cdot 2 \cdot 1 = 8$,

$$|S| = f(1) = L_5(1)\, q(1) = 8\, q(1) \geq 32.$$

Equality requires $(q_1, q_2, q_3) = (1, 2, 1)$, i.e. $q = x(1 + x)^2$, giving exactly the multiset

$$f = x\prod_{i=1}^{5}(1+x^i) = \{1, 2, 3, 3, 4, 4, 5, 5, 5, 6, 6, 6, 7, 7, 7, 8, 8, 8, 9, 9, 9, 10, 10, 10, 11, 11, 12, 12, 13, 14, 15, 16\},$$

which has index 5, maximum element 16, and 32 elements. All constraints are met, so 32 is achievable and minimal.

0.7 Conclusion

$$N = 5, \qquad M = 16, \qquad \min|S| = 32.$$

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